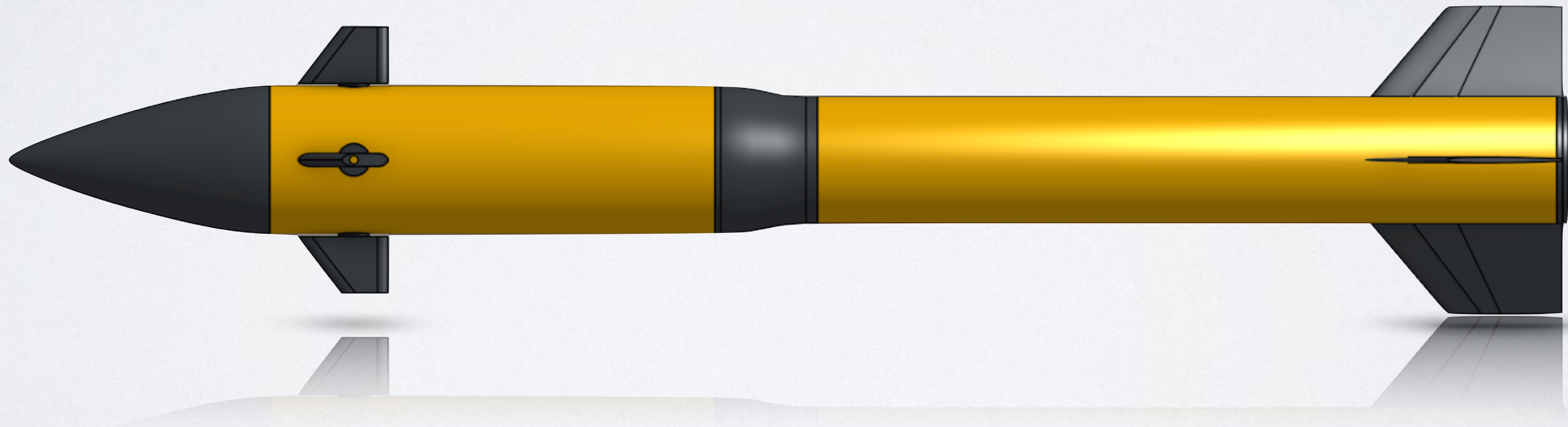


MIRA COSTA HIGH SCHOOL AEROSPACE CLUB

Mira Costa High School / TEAM 25-1126



Noah Weiss
Design Lead

Tesla Pratt
Electrical Lead

Carolina Hodgeon
Programming Lead

Luc Kolczak
Recovery Systems

Laszlo Godec
Design Team

John Tian
Programming Team

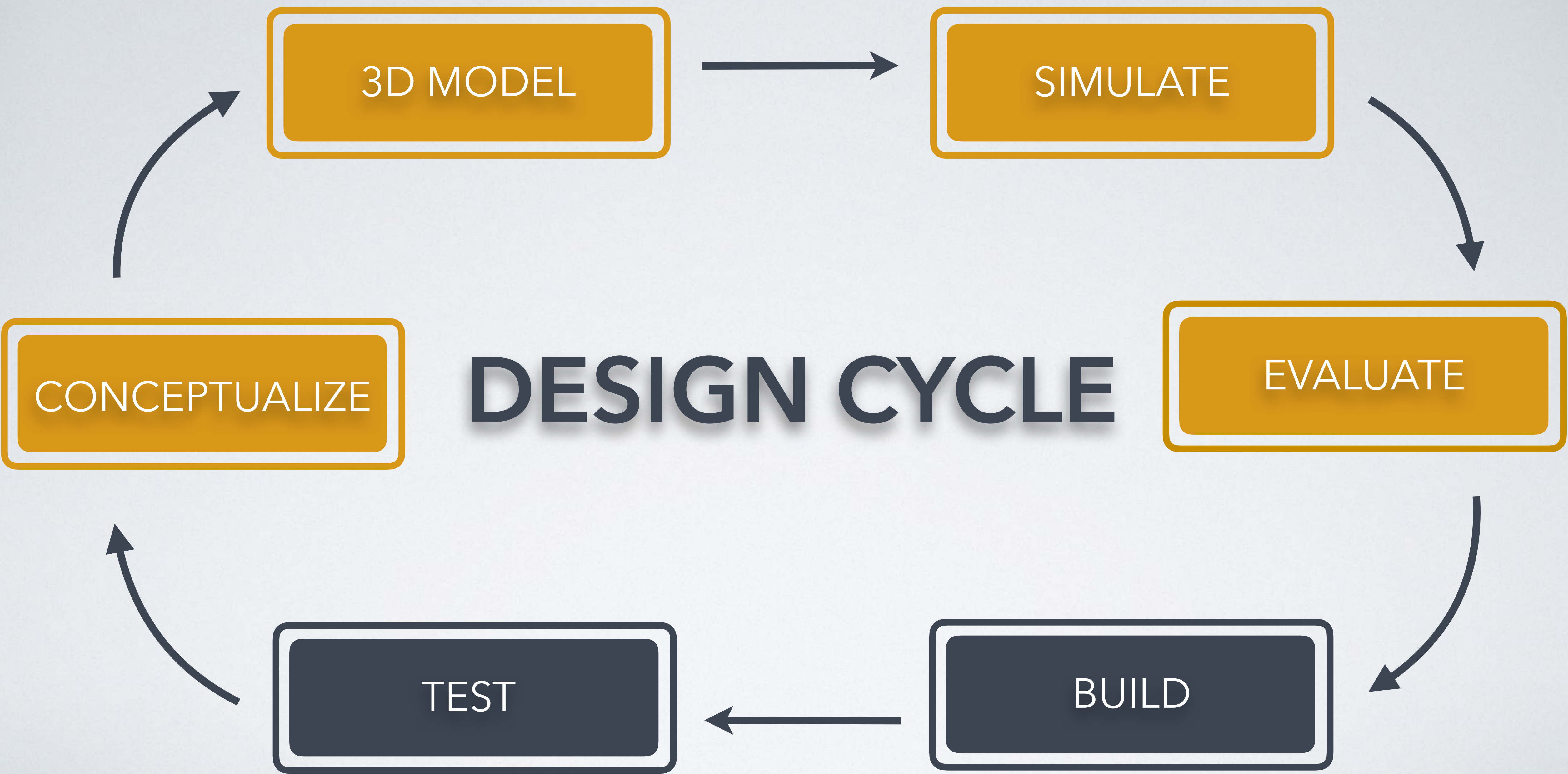
Atticus O'Neill
Design Team

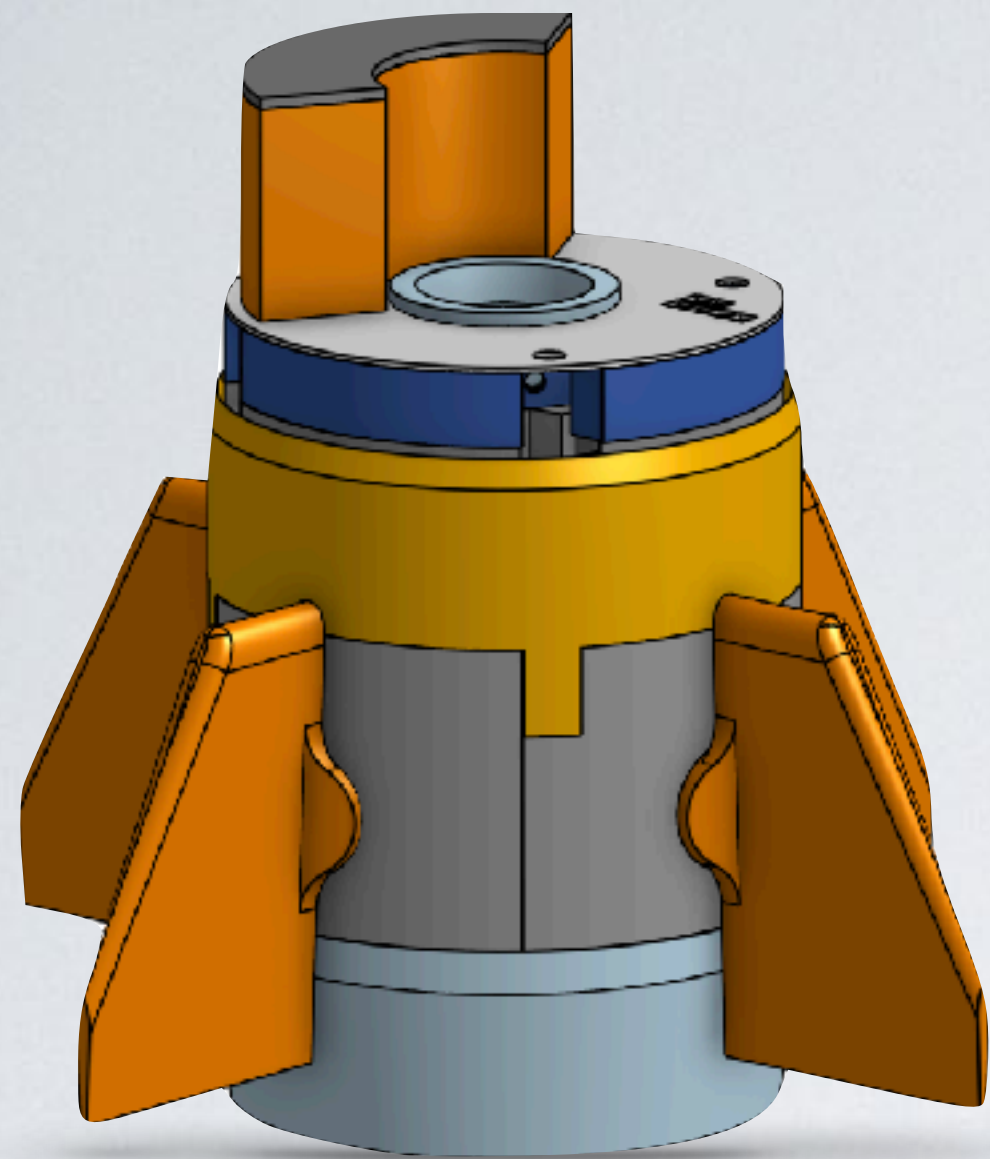
Ayaan Aggarwal
Launch Systems

Saige Beising
Electrical Team

Shane Sheth
Launch Systems



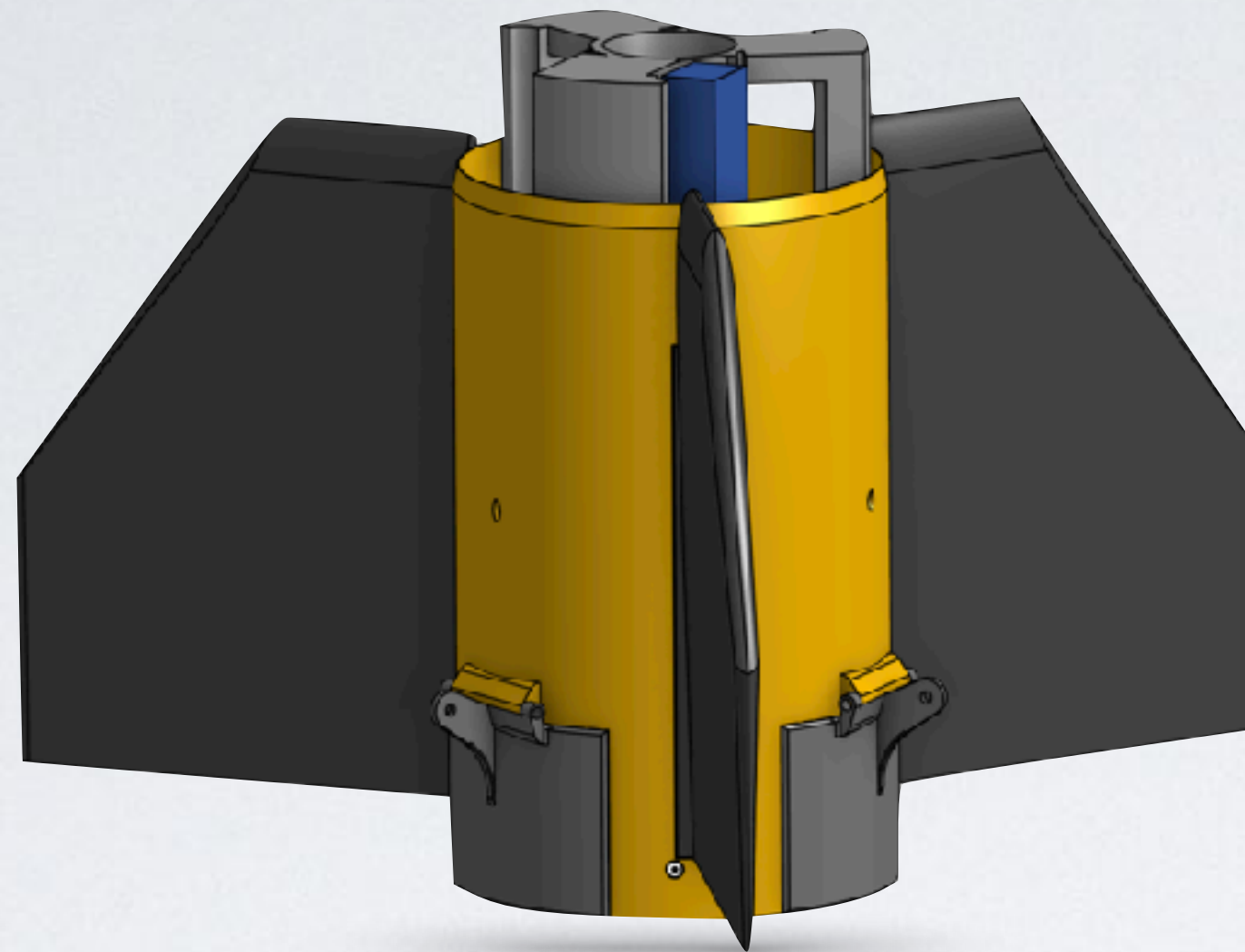




HELIOS I

Aft Controlled Fins Design Principle

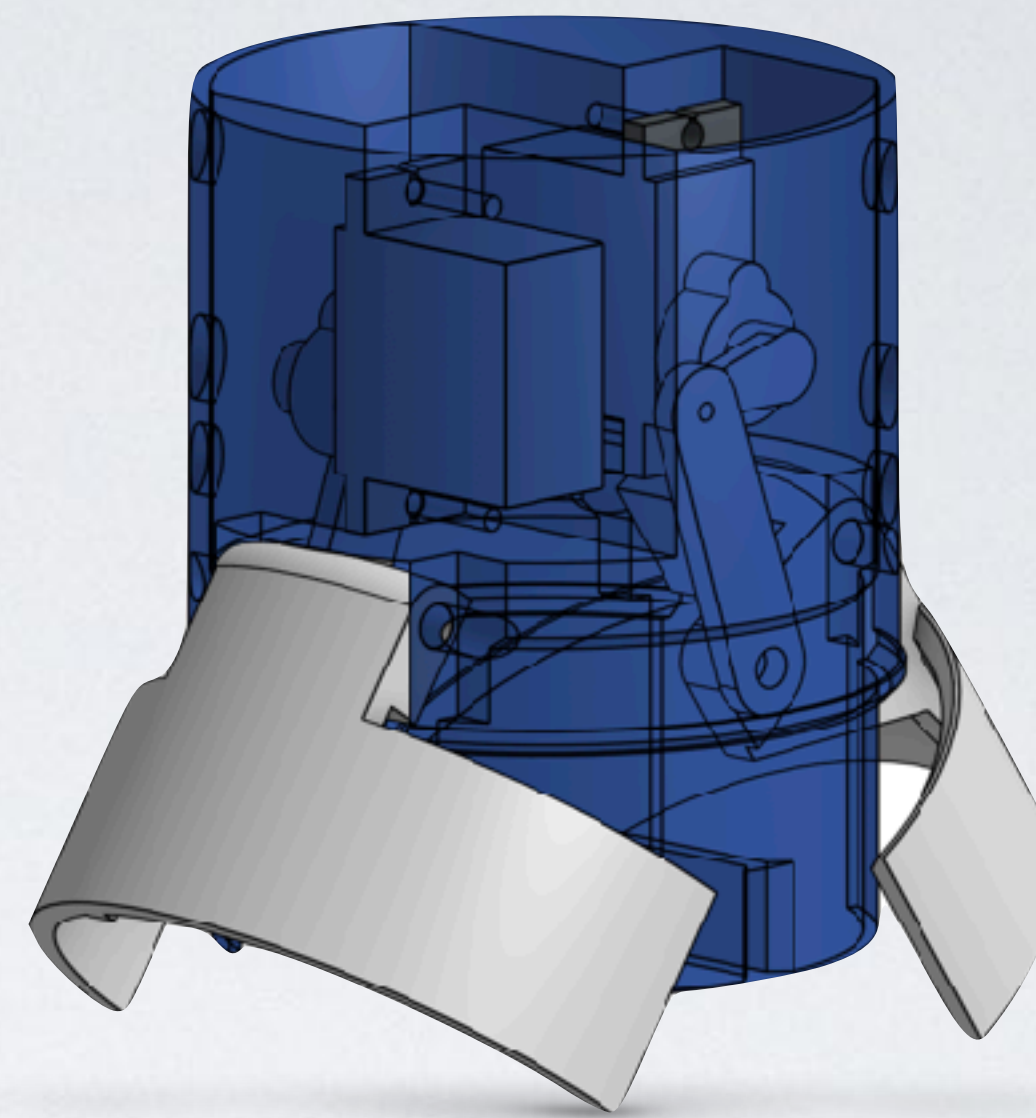
Aft mounted fin system with four high precision 5.4g servos. Constrained between motor outer diameter and tubing inner diameter.



STRATA I

Airbrake Controlled Design Principle

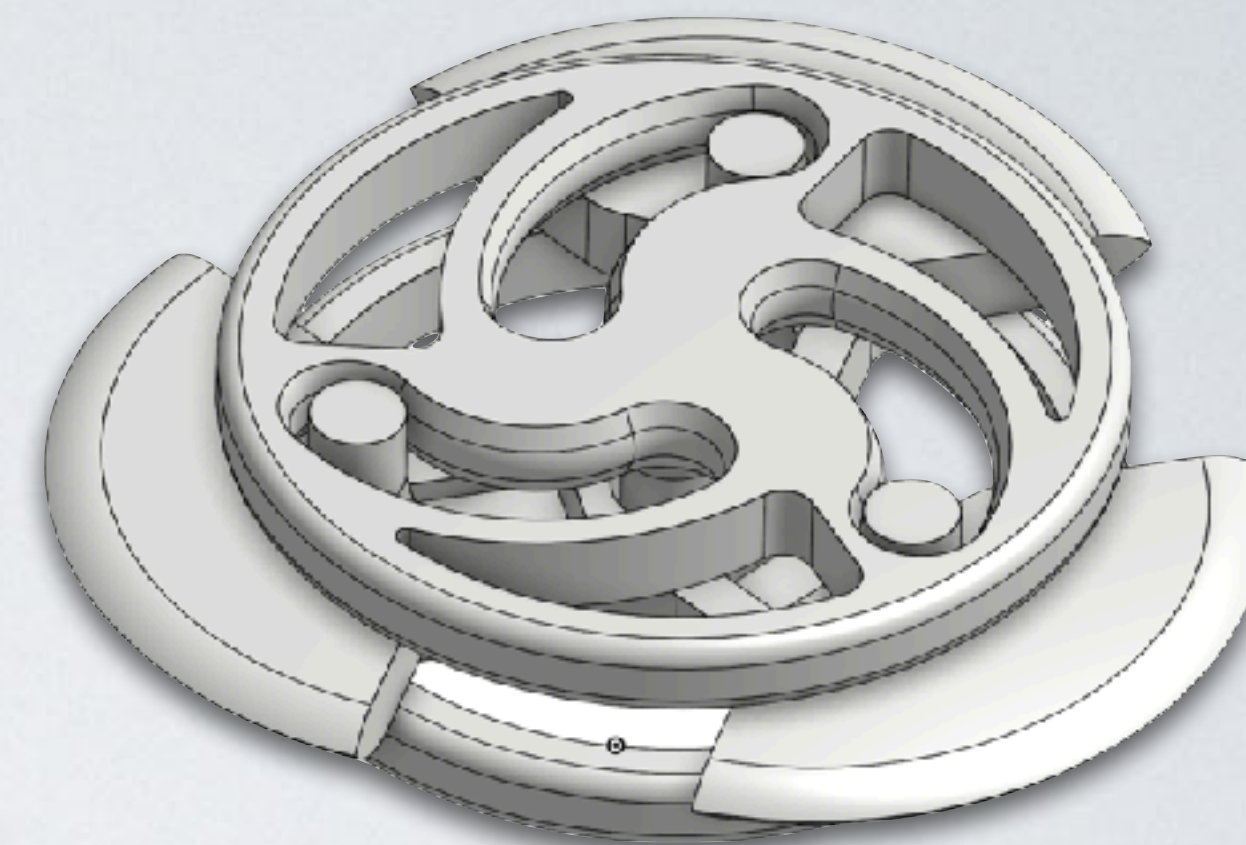
Aft mounted airbrake system with three independently controlled variable drag flaps.



STRATA II

Airbrake Controlled Design Principle

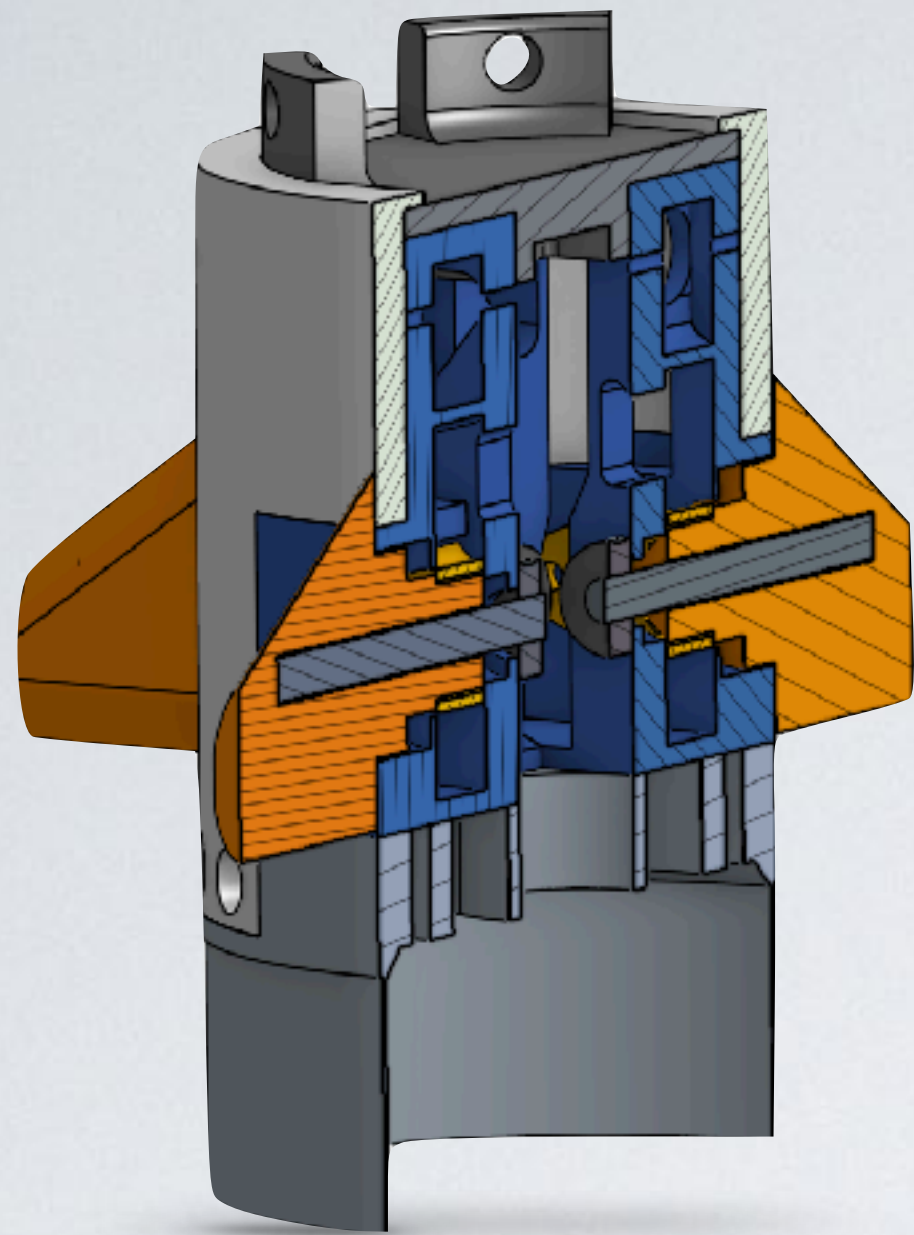
Forward mounted airbrake system with two independently controlled variable drag flaps.



STRATA III

Airbrake Controlled Design Principle

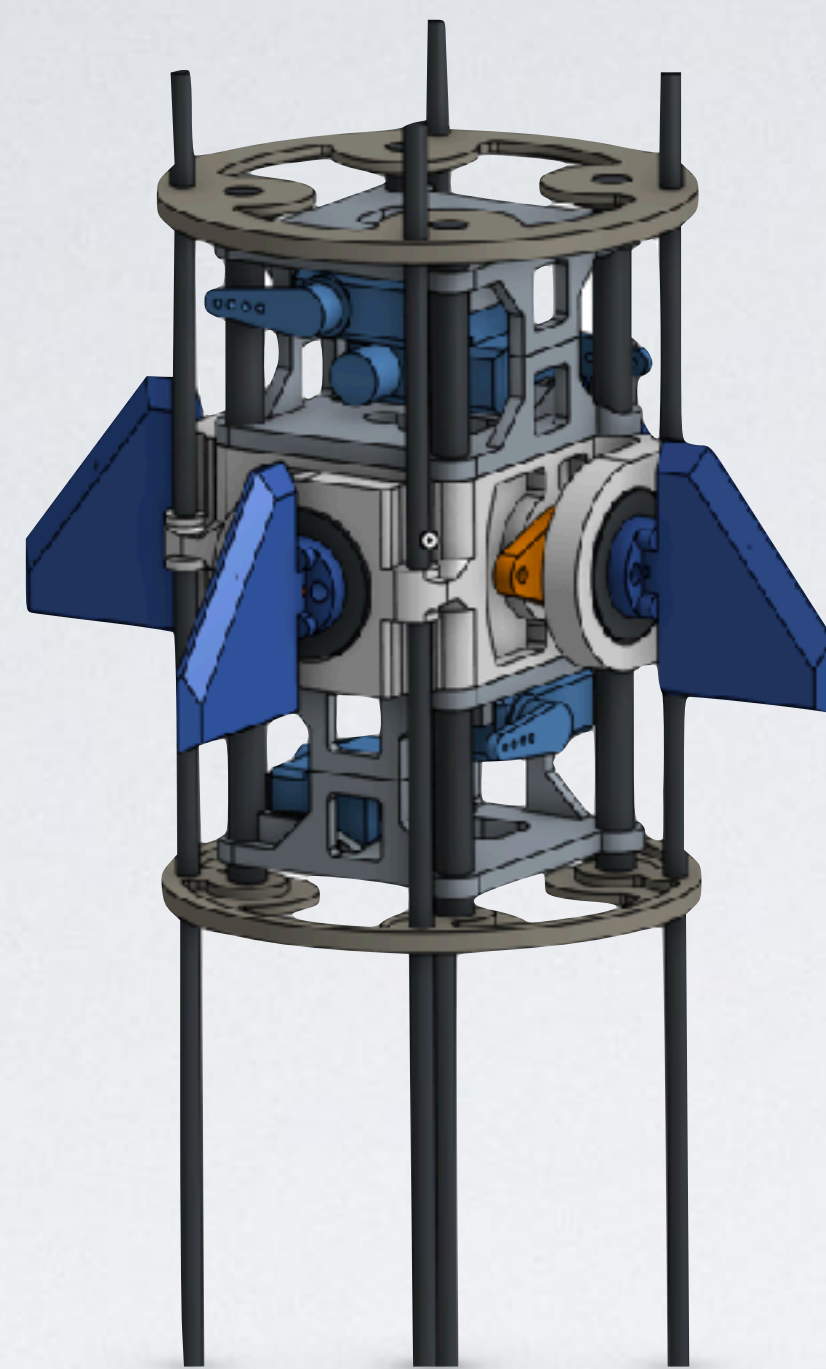
Forward mounted airbrake system with three mechanically connected drag slates controlled by a rotating upper plate.



HELIOS II

Canard Controlled Fins Design Principle

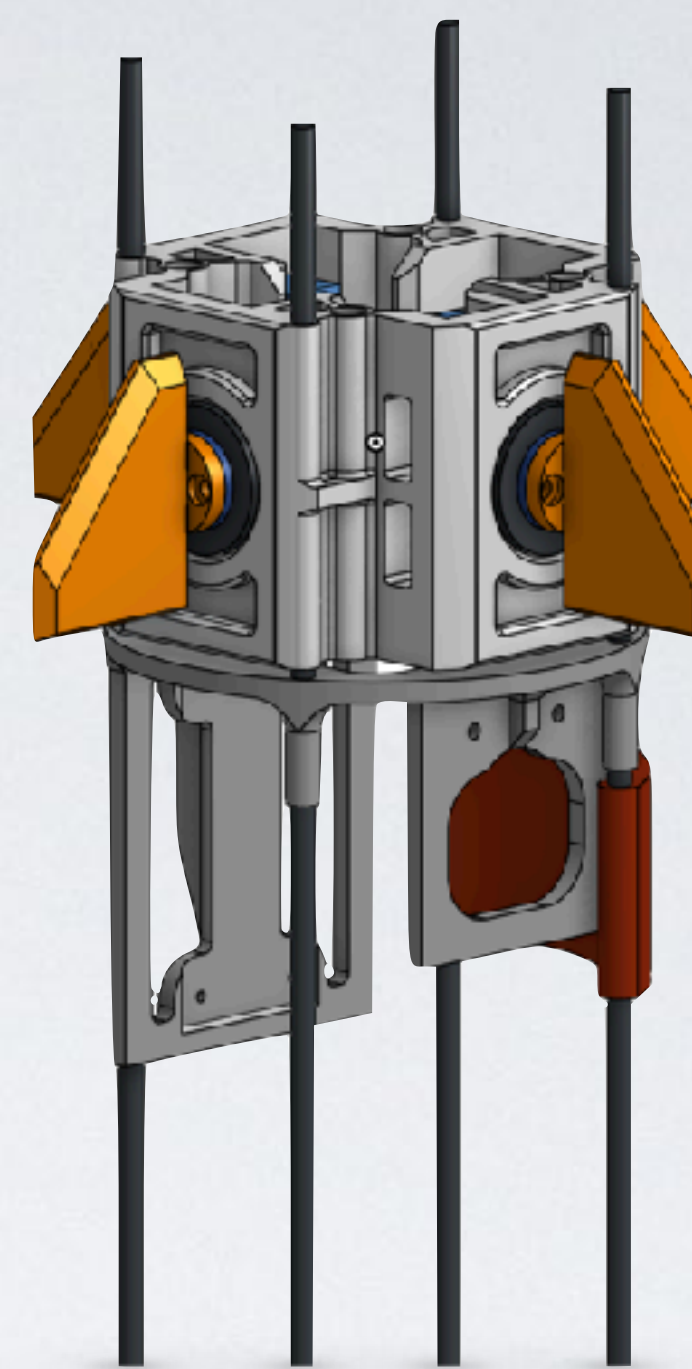
Forward mounted fin system with integrated nosecone mount and constrained only to tubing inner diameter.



HELIOS III

Canard Controlled Fins Design Principle

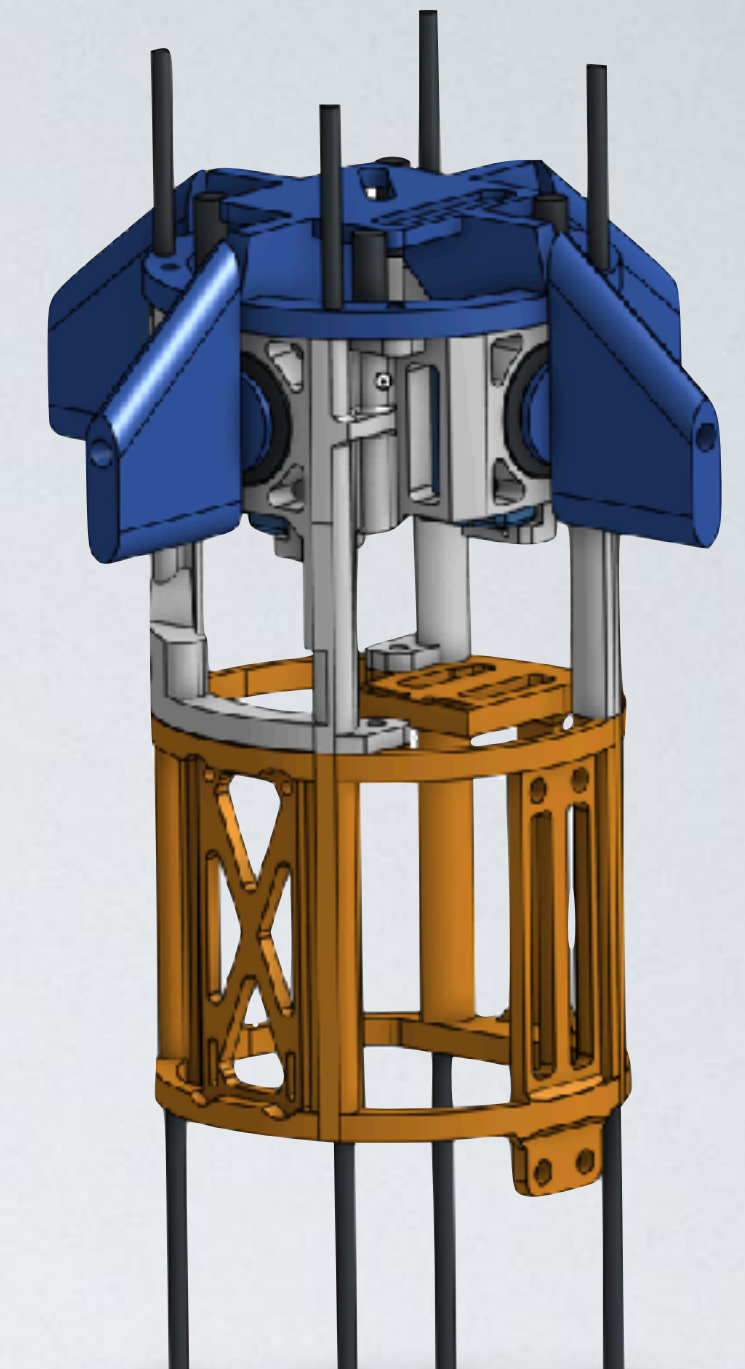
Forward mounted fin system with new carbon rod mounting system and updated aero placement.



HELIOS IV

Canard Controlled Fins Design Principle

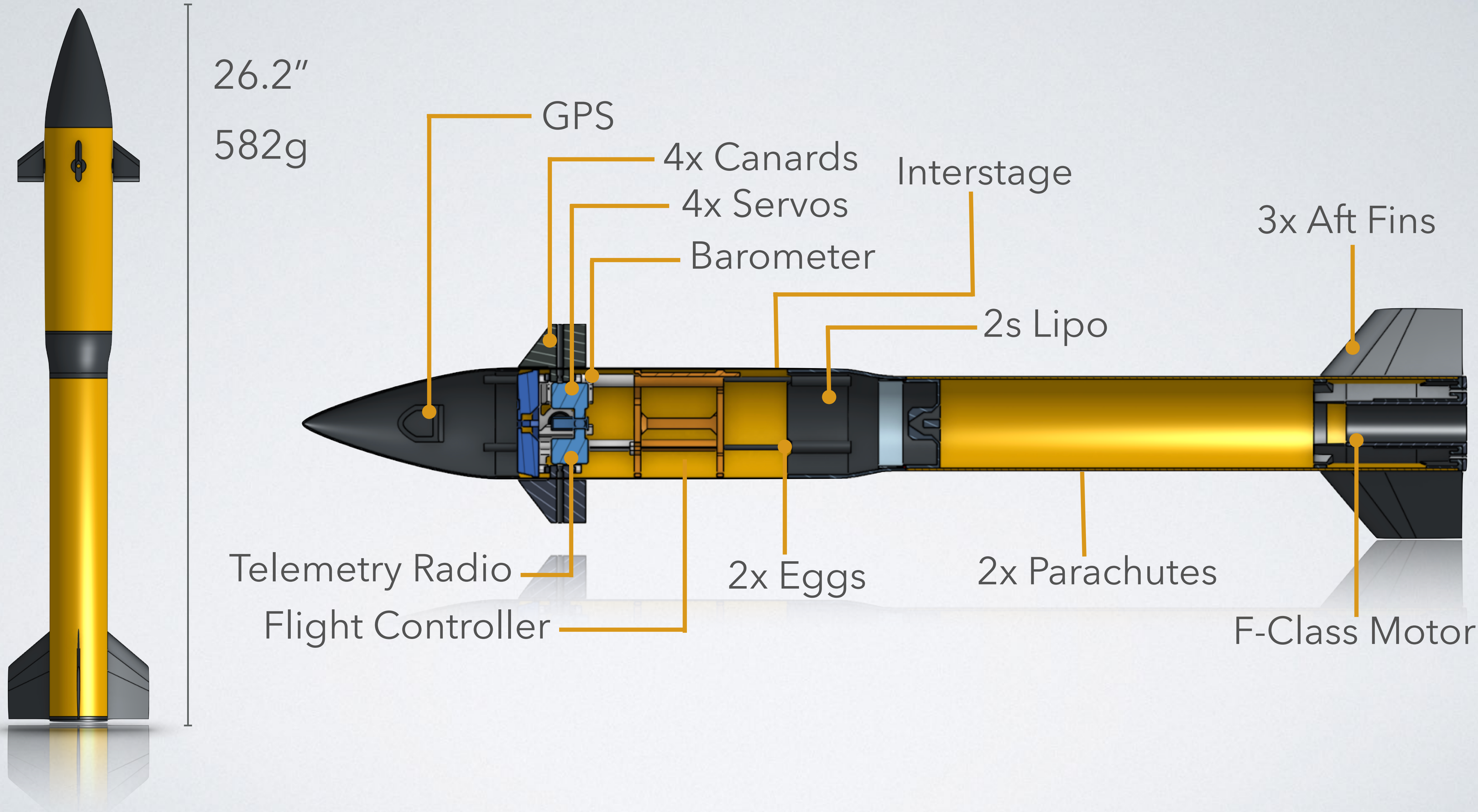
Forward mounted fin system with reduced no. of servos for less weight+greater efficiency.



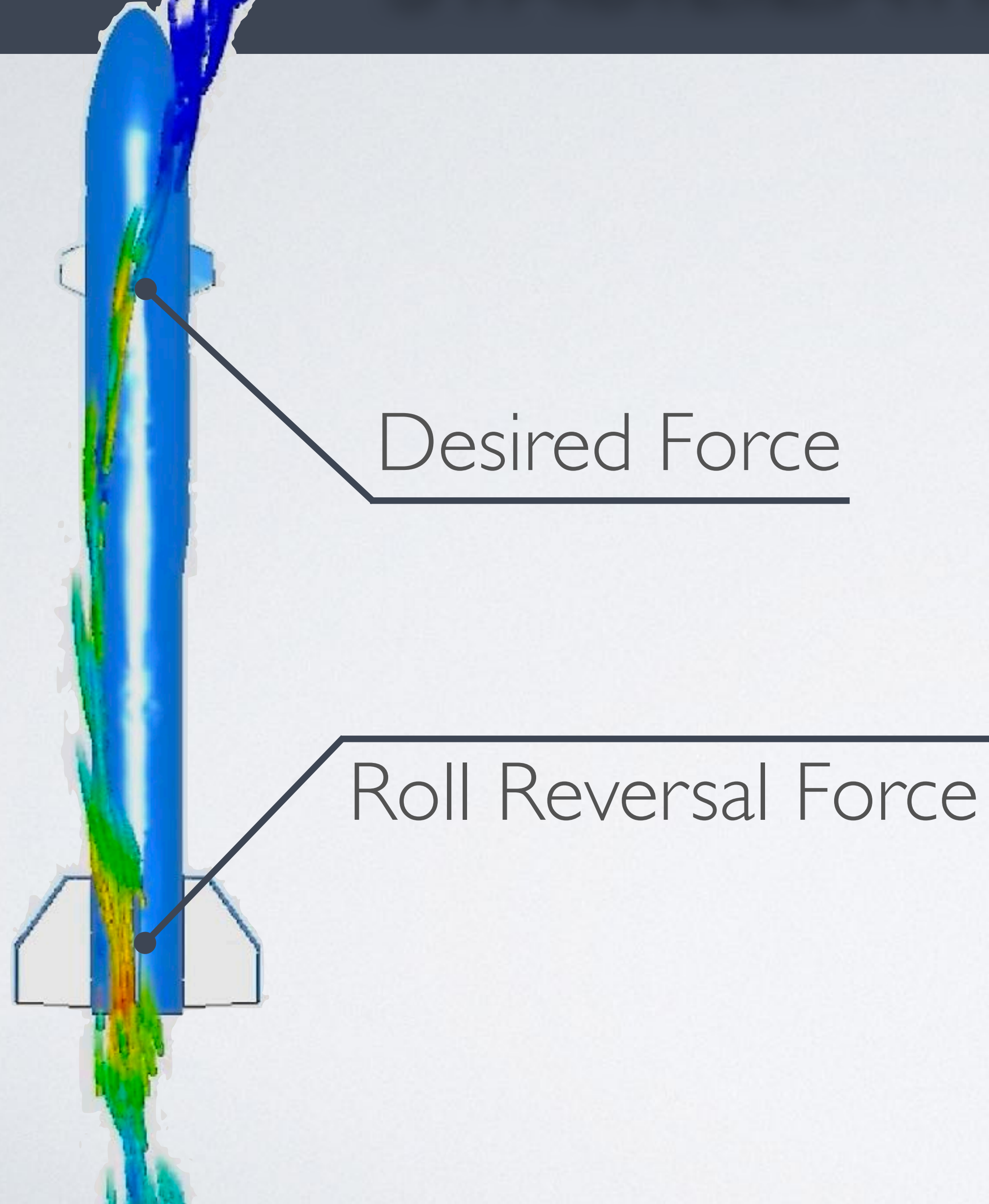
HELIOS V

Canard Controlled Fins Design Principle

Forward mounted fin system with updated servo placement and lightened design. Allowing mounting of more sensors.



STABILIZATION ANALYSIS



*Graphic above shows airflow path over
The canards and into the aft fins,
causing them to rotate*

Ideas

Pros

Cons

Spin Fins

Gyroscopic
stabilization from
roll forces

Causes gyroscopic
precession, and causes
adverse aerodynamic
effects

Vortex
Generators

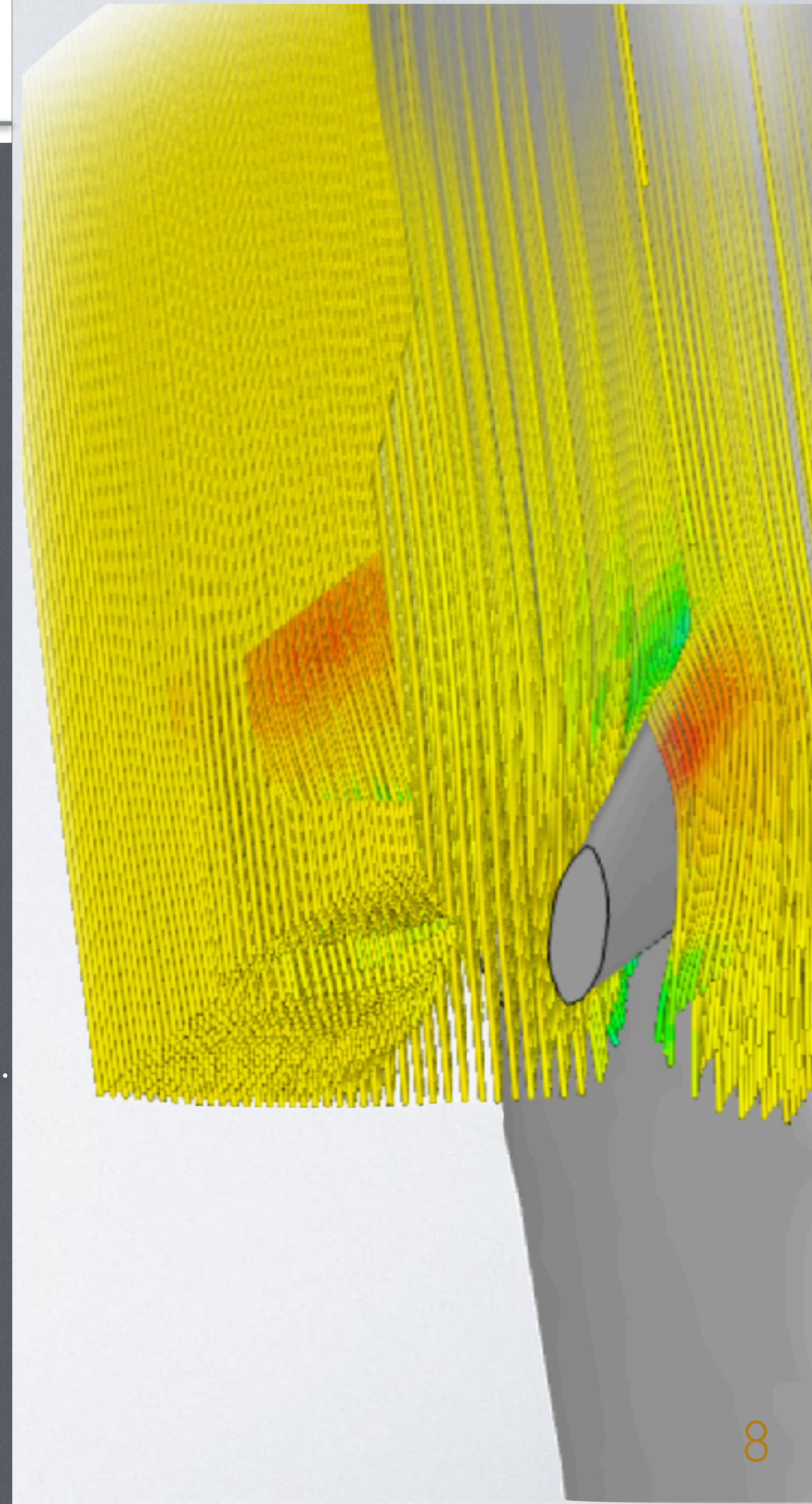
Lightweight, little
mechanical
interference

Mechanically
complicated, can cause
adverse aerodynamic
effects

Programming

No additional weight

Can limit
controllable range



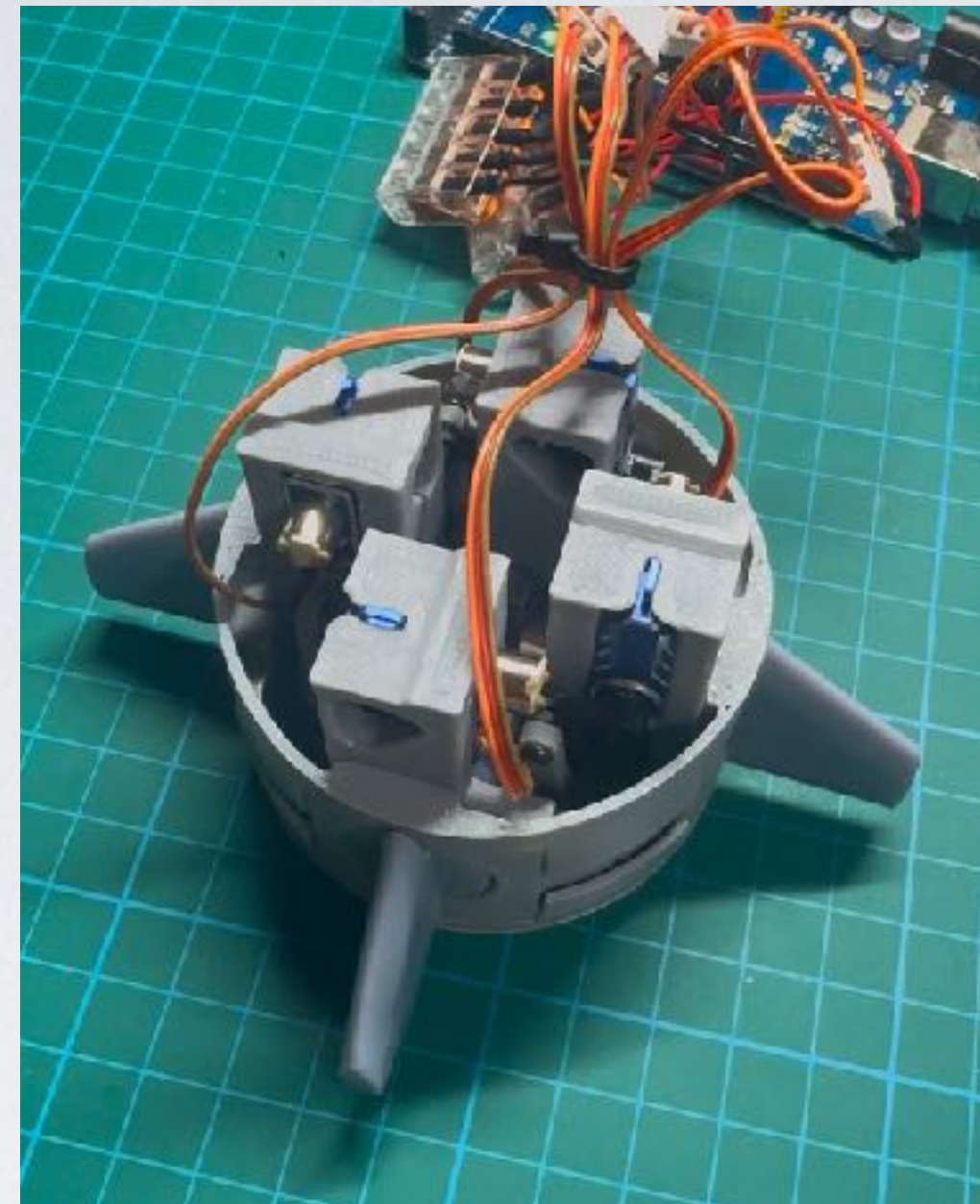


HELIOS I

Aft Controlled Fins *Design Principle*

*Pushrod-Servo powered
aft fin guidance
package.*

Mass: 192g



HELIOS II

Canard Controlled Fins *Design Principle*

*Pushrod-Servo powered
canard fin guidance
package.*

Mass: 172g

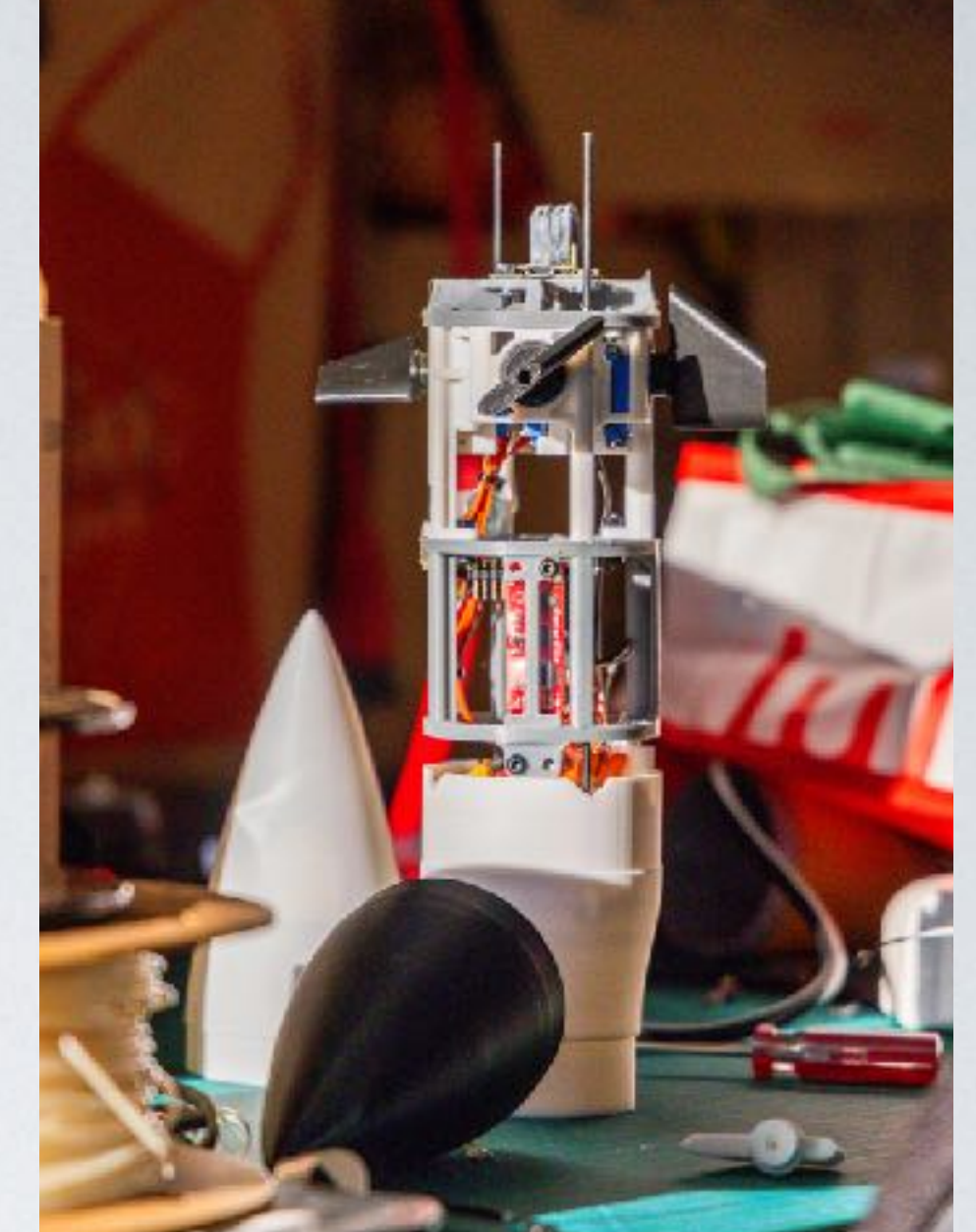


STRATA II

Air Brake Controlled Fins *Design Principle*

*Pushrod-Servo powered
airbrake brake package.*

Mass: 92g



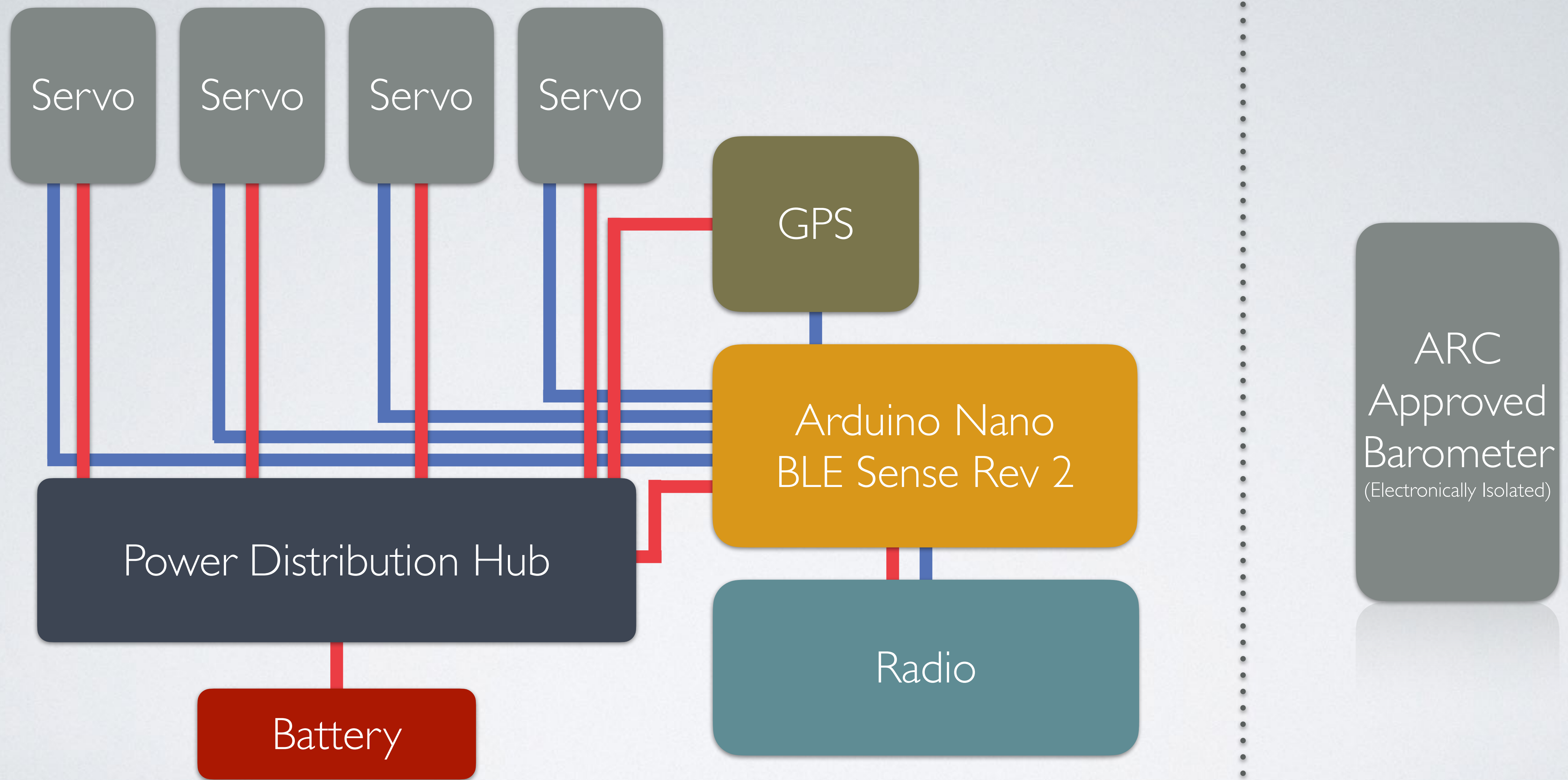
HELIOS IV

Canard Controlled Fins *Design Principle*

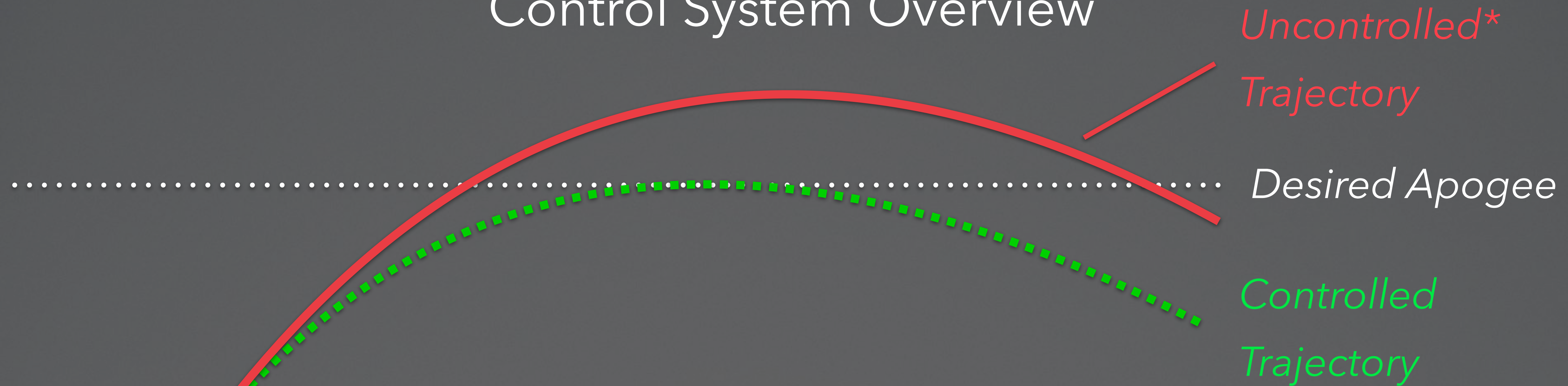
*Pushrod-Servo powered
canard fin guidance
package.*

TOTAL Mass: 210g

Canard Portion: 82g



Control System Overview



**Uncontrolled trajectory is inclusive of canard controlled stabilization.
Meaning it assumes the rocket is pointed upwards through the flight and
accounts for predicted drag produced by small canard movements.*



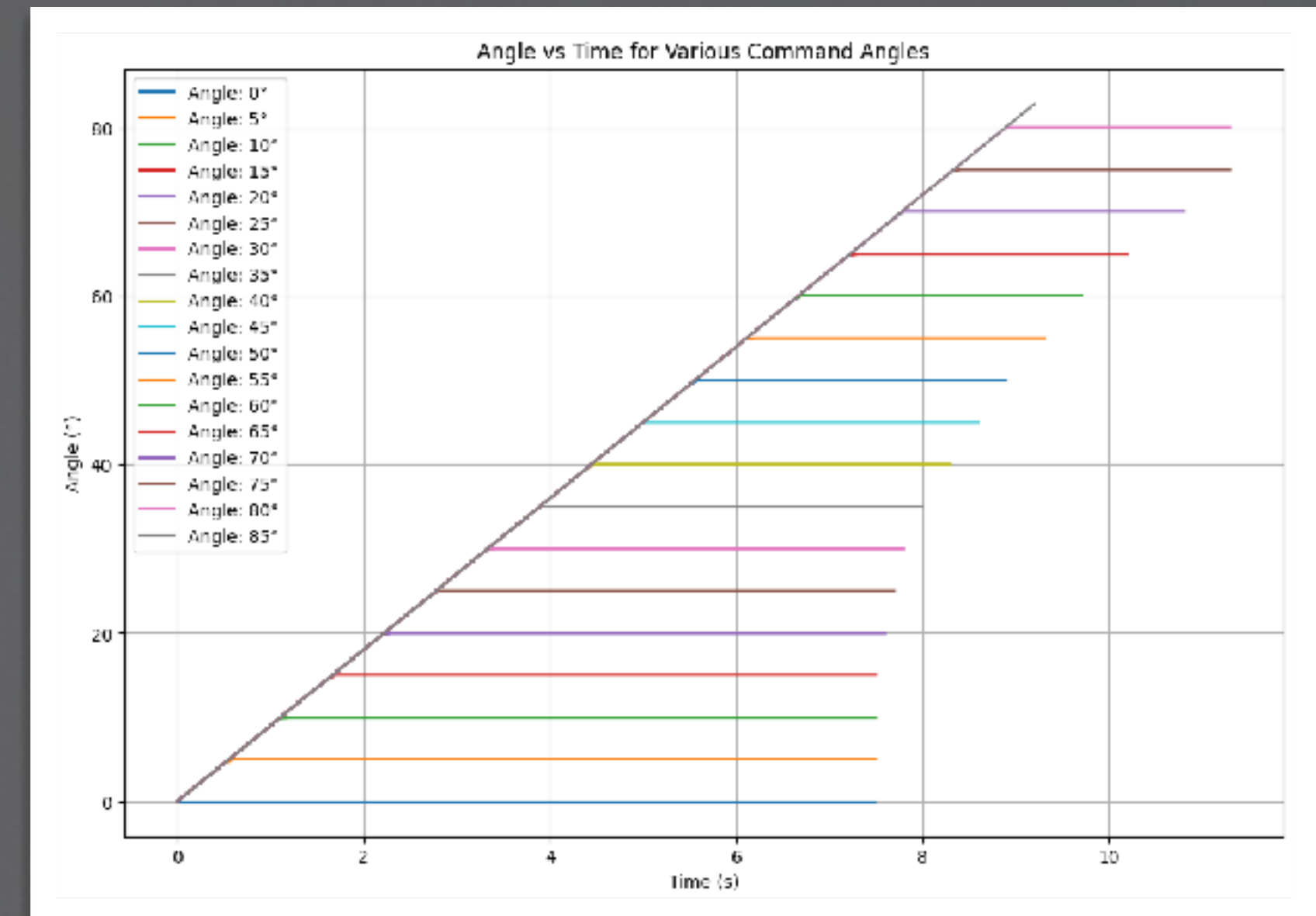
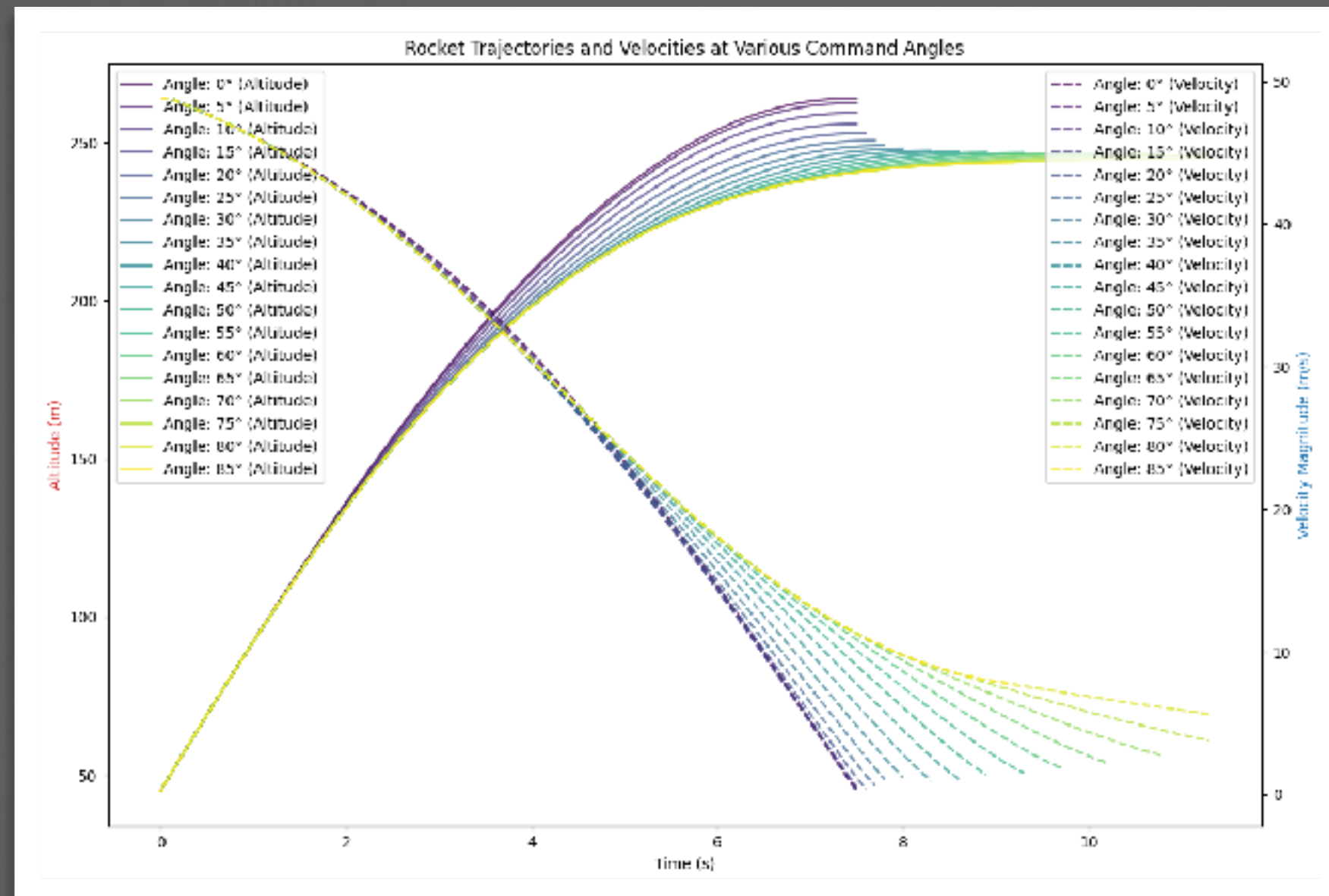
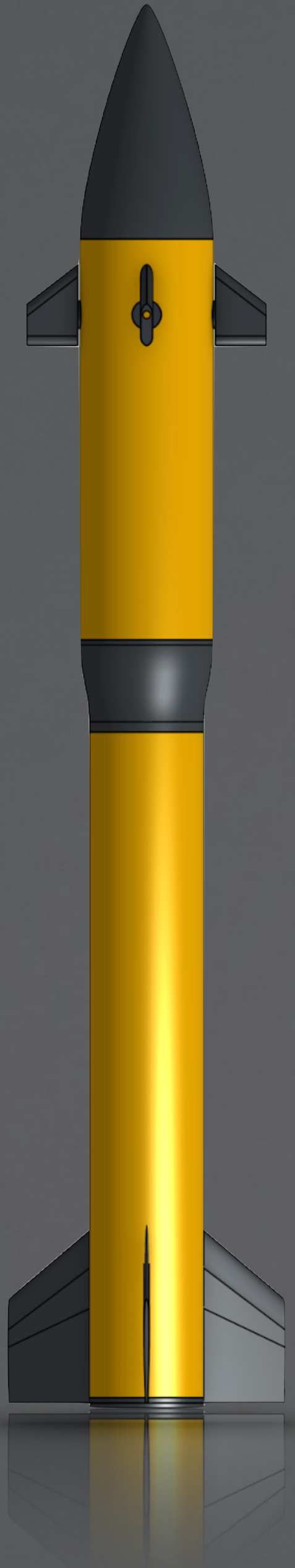
Drag Calculations

Air Density with respect to Altitude

Cross Sectional Area vs. Canard Angles

$$F_{\text{Drag}} = \frac{1}{2} \cdot \left(\frac{P_0 \cdot \left(1 - \frac{Lh}{T_0}\right)^{5.255}}{R \cdot (T_0 - Lh)} \right) \cdot v^2 \cdot C_D(t) \cdot \left(\pi r^2 + \ell_{\text{canard}} \cdot (\sin(\theta_1) + \sin(\theta_2) + \sin(\theta_3) + \sin(\theta_4)) \right)$$

Coefficient of Drag determined experimentally as a function of speed



PID Control

*proportional gain: strength of
response to current error*

*derivative of error:
reduces oscillation*

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de}{dt}$$

PID Variable

*integral gain: strength of
response to error over time*

```
for (int i = 0; i < samples; i++) {
  if (IMU.gyroscopeAvailable() && IMU.accelerationAvailable() && IMU.magneticFieldAvailable()){
    IMU.readGyroscope(gx, gy, gz);
    IMU.readAcceleration(ax, ay, az);
    IMU.readMagneticField(mx, my, mz);
    gx -= gyroXOffset;
    gy -= gyroYOffset;
    gz -= gyroZOffset;
    updateMag(mx, my, mz);
    applyScalingFactor(mx, ny, nz, M_scale);
    applyMagneticDeclination(mx, my, -0.19); // local magnetic declination
    // Convert Accel from g's to m/s^2
    ax *= 9.80665; ay *= 9.80665; az *= 9.80665;
    // Convert Gyro from deg/s to rad/s
    gx *= PI / 180.0; gy *= PI / 180.0; gz *= PI / 180.0;
    deltat = fusion.deltatUpdate();
    // Correct axes for sensor fusion library (-gx, gy, gz, -ax, ay, az)
    fusion.MadgwickUpdate(-gx, gy, gz, -ax, ay, az, mx, ny, mz, deltat);
    sum_pitch += fusion.getPitch();
    sum_roll += fusion.getRoll();
    sum_yaw += fusion.getYaw();
    actual_samples++;
    delay(10);
  } else {
    // If sensors aren't available briefly, just skip sample
    delay(10);
  }
}
```

```
// roll PID
float error_roll = desired_roll - current_roll;
if (fabs(error_roll) < deadband) error_roll = 0;
roll_integral += error_roll * deltat;
// roll_integral = constrain(roll_integral, -INTEGRAL_LIMIT, INTEGRAL_LIMIT);
float d_roll = (error_roll - roll_prev_error) / deltat;
float roll_output = Kp_roll * error_roll + Ki_roll * roll_integral + Kd_roll * d_roll;
roll_prev_error = error_roll;

// yaw PID
float yaw_rate = (current_yaw - yaw_prev) / deltat; // calculate yaw rate
float error_yaw = desired_yaw_rate - yaw_rate; // error is difference from desired rate aka 0
if (fabs(error_yaw * 180.0/PI) < deadband) error_yaw = 0; // Apply deadband in deg/s
yaw_integral += error_yaw * deltat;
// yaw_integral = constrain(yaw_integral, -INTEGRAL_LIMIT, INTEGRAL_LIMIT);
float d_yaw = (error_yaw - yaw_prev_error) / deltat;
float yaw_output = Kp_yaw * error_yaw + Ki_yaw * yaw_integral + Kd_yaw * d_yaw;
yaw_prev_error = error_yaw;
yaw_prev = current_yaw; // Update previous yaw for next rate calculation
```


LAUNCH DAY PROCEDURE

Upload Code

```
the gyro is stable
< maxDrift && abs(avgY) < maxDrift && abs(avgZ) < maxDrift
ln("✅ Gyro is stable and ready for launch.");
t("Avg X: "); Serial.print(avgX);
t(" | Avg Y: "); Serial.print(avgY);
t(" | Avg Z: "); Serial.println(avgZ);

ln("❌ Gyro is NOT stable. Wait for stability");
t("Avg X: "); Serial.print(avgX);
t(" | Avg Y: "); Serial.print(avgY);
t(" | Avg Z: "); Serial.println(avgZ);
e;
```

Secure Assembly



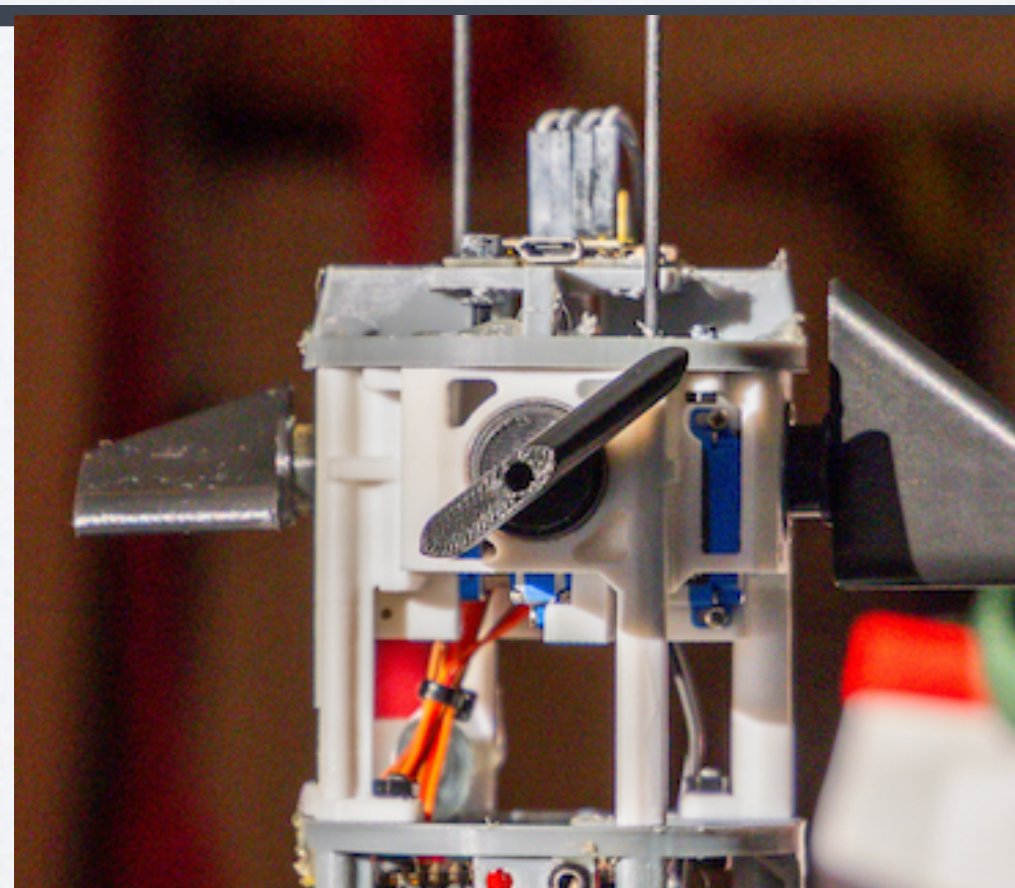
Position Rocket



Ignite Motor



Full Calibration



Bluetooth Connect



FLIGHT TEST 1



RESULT: Guidance System Failure
MAX ALTITUDE: 382 ft

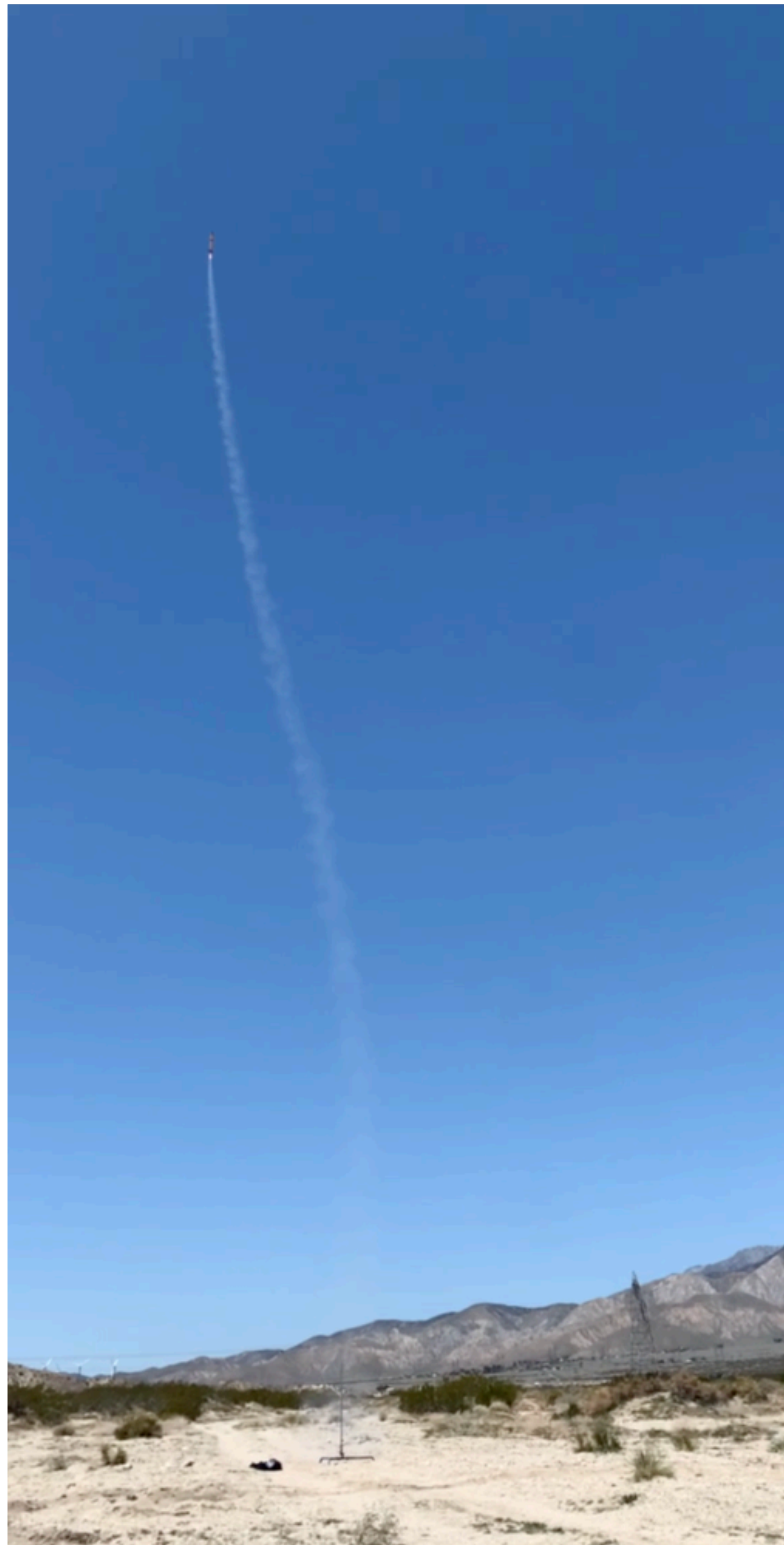
Approximately 1.2 seconds into flight, two fins became disconnected leading to complete loss of control and a rapid acceleration on the pitch axis.

FLIGHT TEST 2

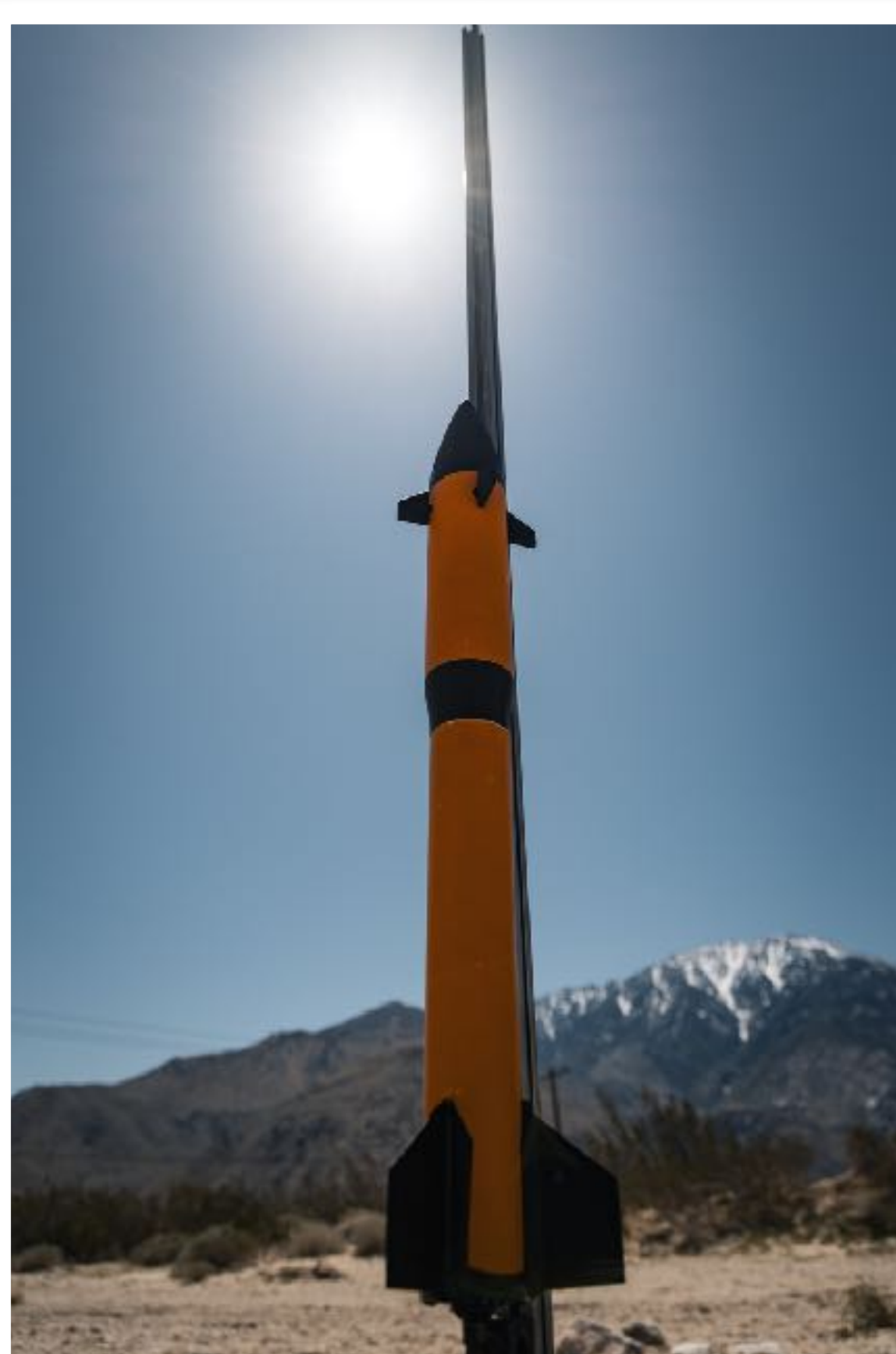
RESULT: Successful ascent + recovery failure
MAX ALTITUDE: 792 ft

The rocket achieved some of its flight requirements:

- *Success: Orientation Correction*
- *Success: Altitude Control*
- *Failure: Parachute Deployment*
- *Failure: Recovery Systems*



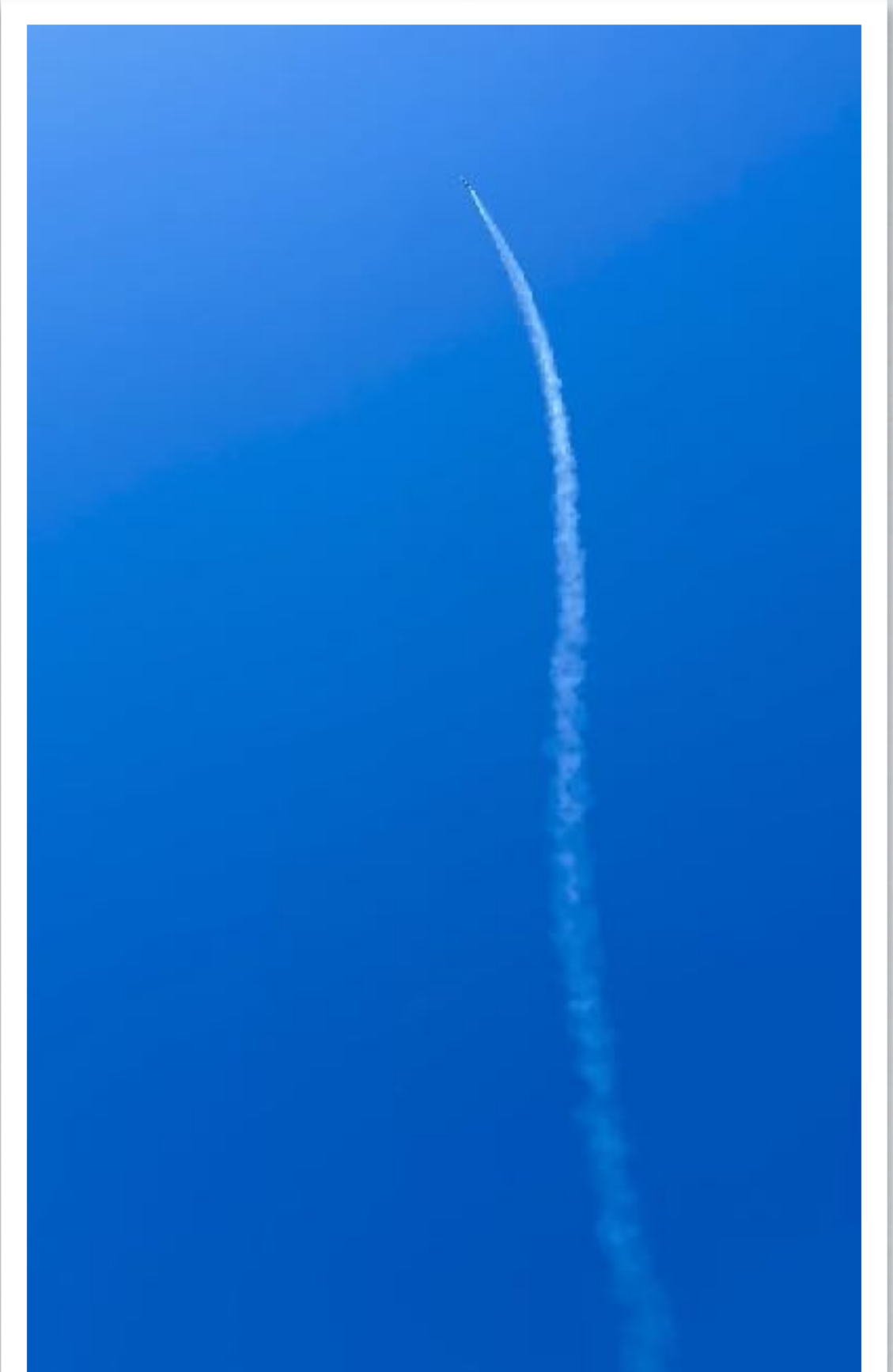
FLIGHT TEST 3



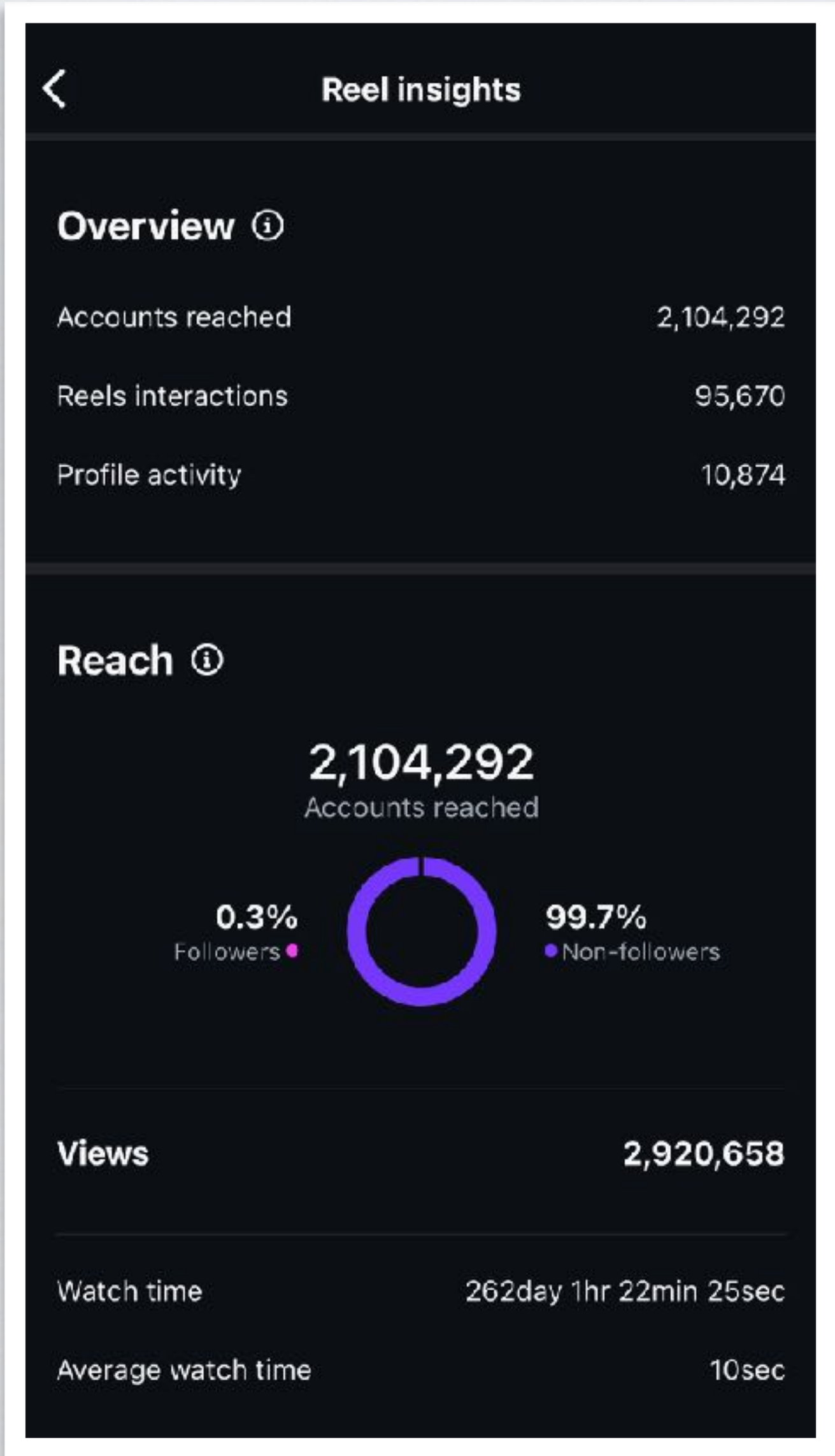
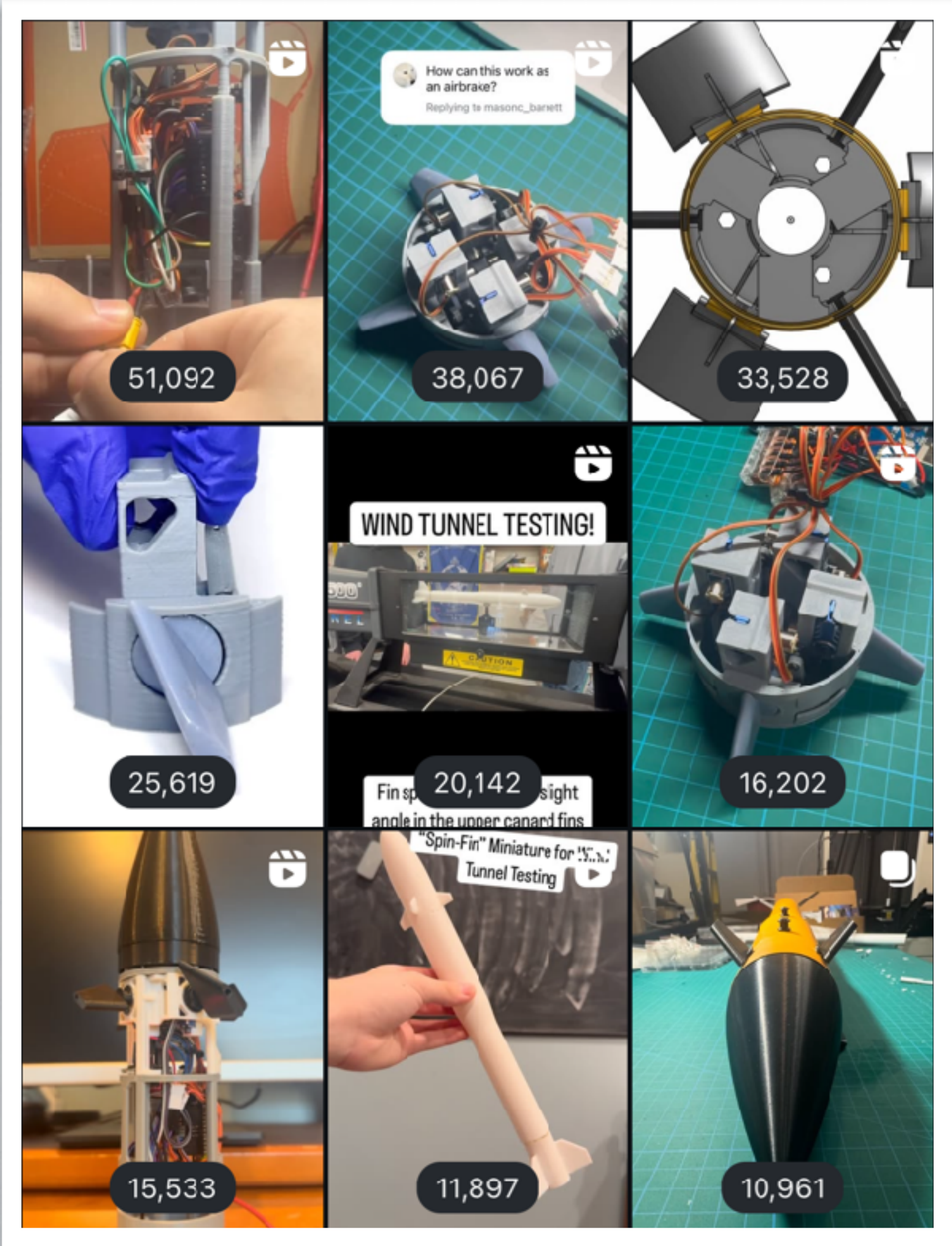
RESULT: Guidance System Failure
MAX ALTITUDE: 694 ft

The rocket achieved some of its flight requirements:

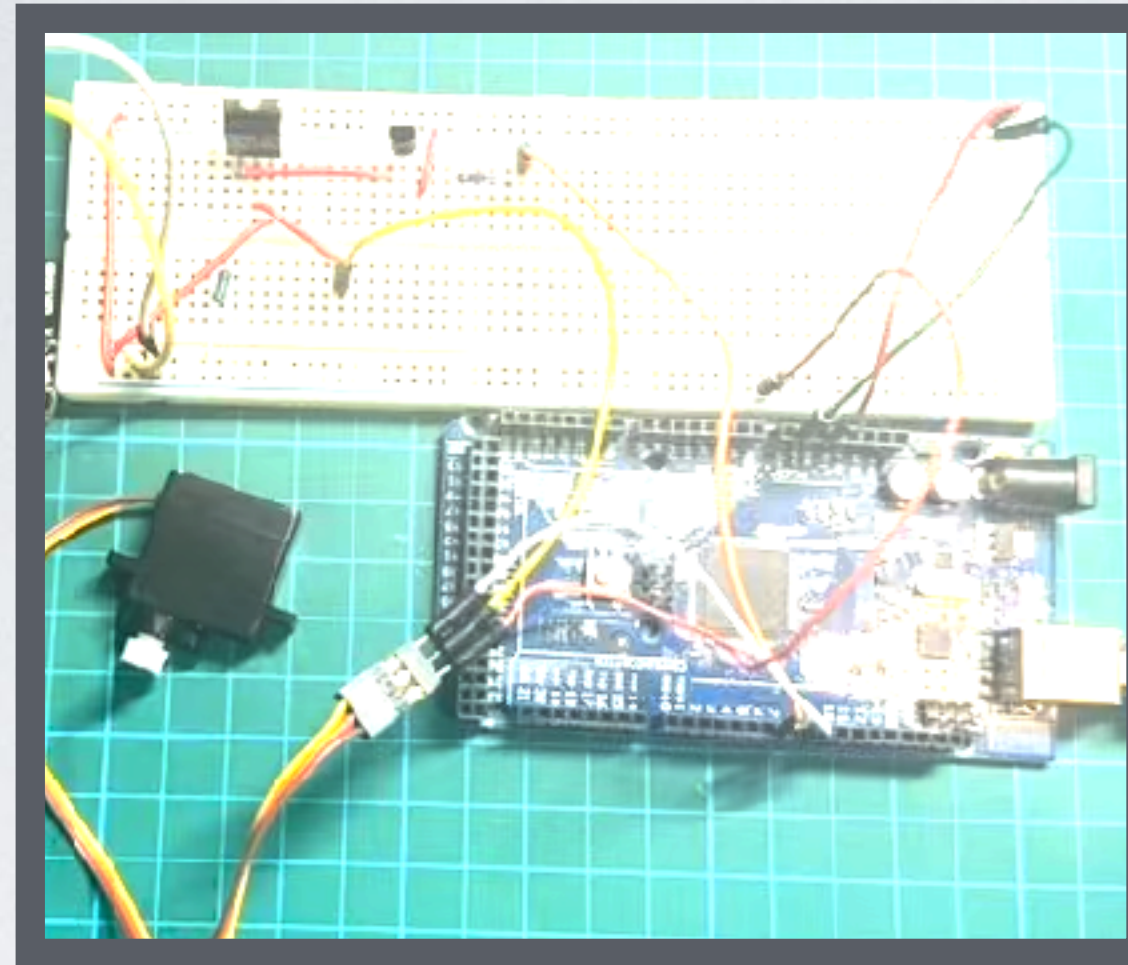
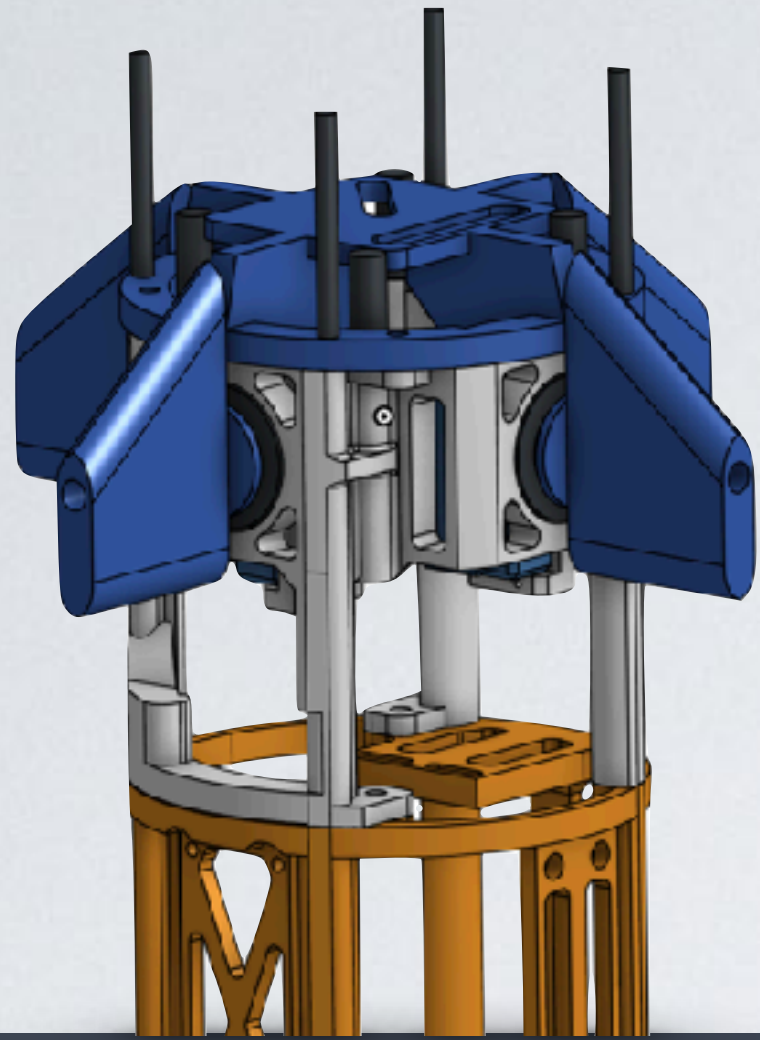
- *Failure: Orientation Correction*
- *Failure: Altitude Control*
- *Success: Parachute Deployment*
- *Failure: Recovery Systems - egg breakage*



Fundraising and Outreach



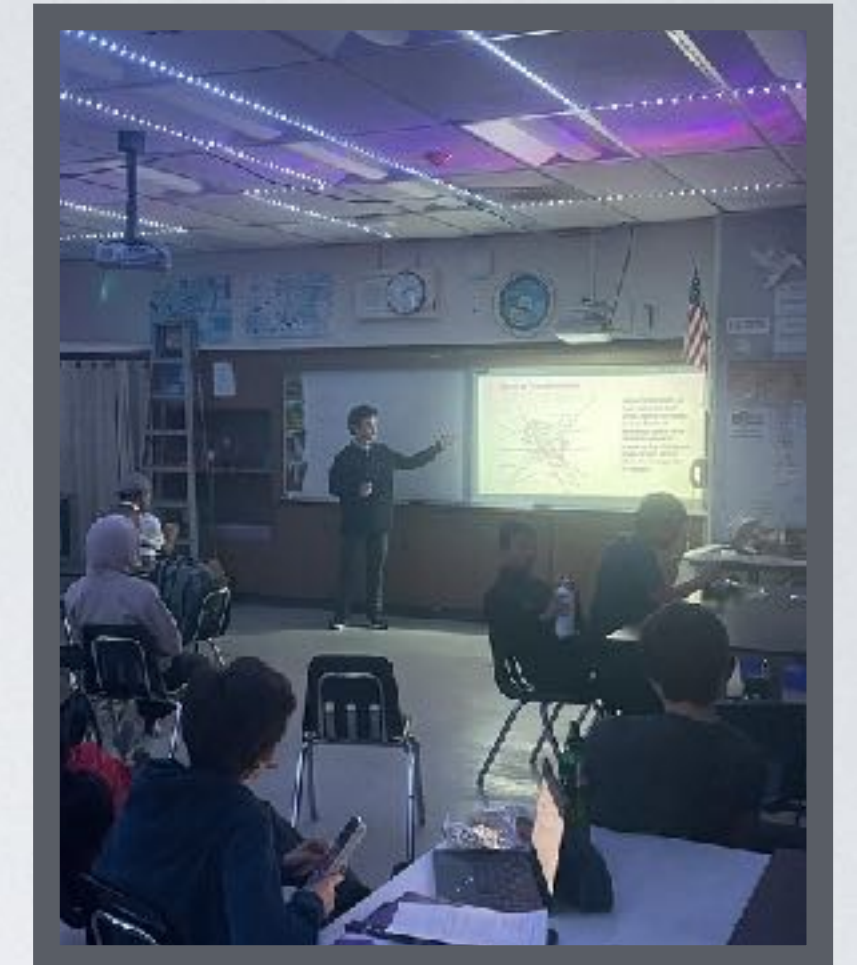
Always Room for Improvement



```
float Kp_pitch = 0.225, Ki_pitch = 1.25, Kd_pitch = 0.0;
float Kp_roll  = 0.225, Ki_roll  = 1.25, Kd_roll  = 0.0;
float Kp_yaw   = 0.1,  Ki_yaw   = 0,  Kd_yaw   = 0.0;

// Setpoints (pointing straight up)
float desired_pitch = 0.0;
float desired_roll  = 0.0;
float desired_yaw_rate = 0.0;

// PID state variables
float pitch_integral = 0, pitch_prev_error = 0;
float roll_integral  = 0, roll_prev_error  = 0;
float yaw_integral   = 0, yaw_prev_error   = 0;
float yaw_prev       = 0;
```



Mechanical

Electrical

Programming

Outreach

Sleeker fins with more reliable attachment

On multiple launch days, our team struggled to attach the canard fins due to the small screw size

Radio hardware for long range communication

Current bluetooth module is unreliable and doesn't work long range

Improved refinements for PID and GYRO control

Dynamically adjusted PID constants according to launch conditions and calibration readings

Involvement in broader STEM communities

Expansion to local middle and elementary schools, fostering early love for rocketry



THANK YOU FOR YOUR TIME

